# Problem G. Gene

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 2.5 seconds |
| Memory limit: | 512 megabytes |

Let $s$ and $t$ be two strings with equal lengths $M$. Define $f(s,t) = \sum_{i=1}^{M} [s_i \neq t_i]$.

You are given $N$ strings $s_1, s_2, \cdots, s_N$ and a constant threshold $K$. Each of the string contains exactly $M$ lowercase letters. You need to perform the following queries $Q$ times:

- Given a string $t$ of length $M$, calculate $\sum_{i=1}^{N} [f(s_i,t) \leq K]$

## Input

The first line of the input contains four integers $N, Q, M$, and $K$ ($1 \leq N, Q \leq 300$, $1 \leq M \leq 60,000$, $1 \leq K \leq 10$).

The $i$-th line of the next $N$ lines contains a single string $s_i$ consisting exactly $M$ lowercase letters.

The $i$-th line of the next $Q$ lines contains a single string $t$ consisting exactly $M$ lowercase letters, indicating a query.

## Output

For each query, output a single line contains a single integer, indicating the answer.

## Examples

| standard input | standard output |
|---|---|
| 6 4 4 1<br>kaki<br>kika<br>manu<br>nana<br>tepu<br>tero<br>kaka<br>mana<br>teri<br>anan | 2<br>2<br>1<br>0 |
| 8 6 7 3<br>delphis<br>aduncus<br>peronii<br>plumbea<br>clymene<br>hectori<br>griseus<br>electra<br>delphis<br>helpiii<br>perphii<br>clumeee<br>eleelea<br>ddlpcus | 1<br>1<br>2<br>2<br>1<br>2 |